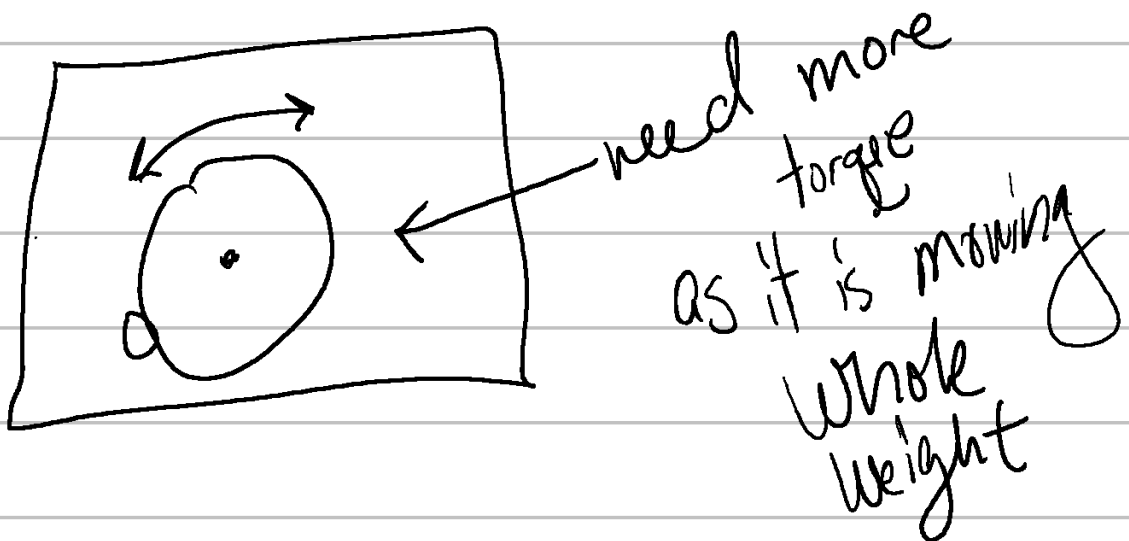
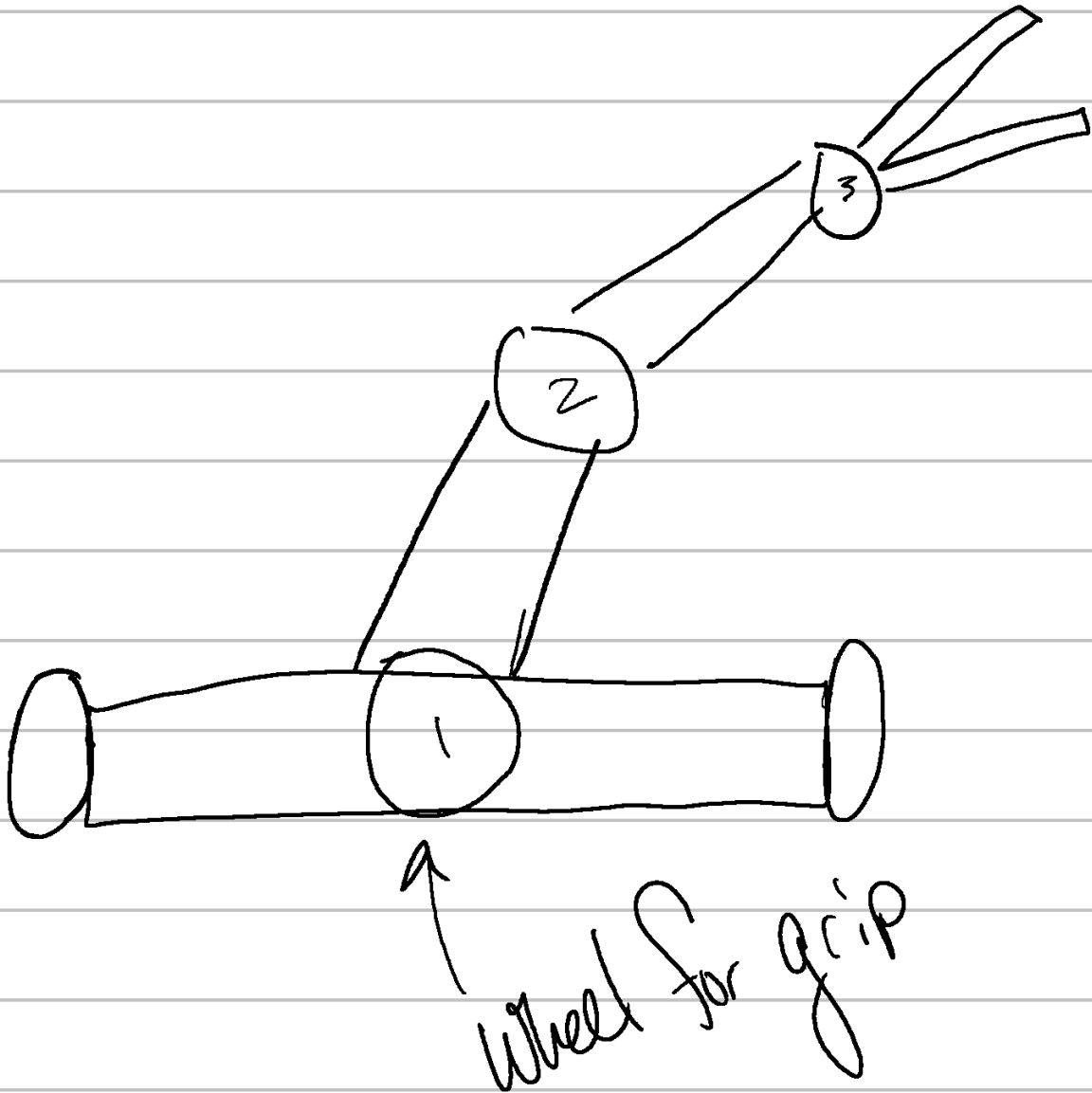
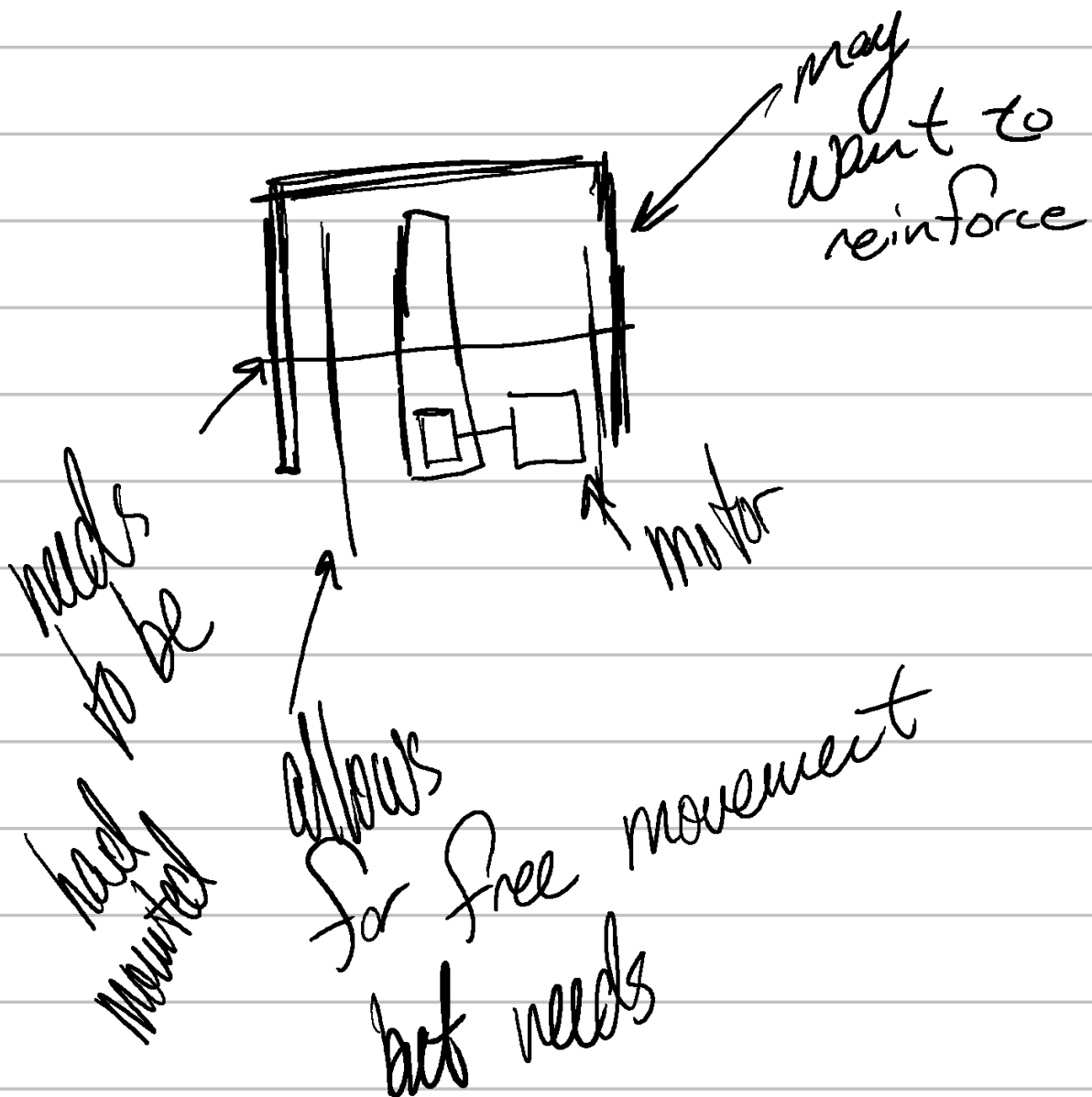
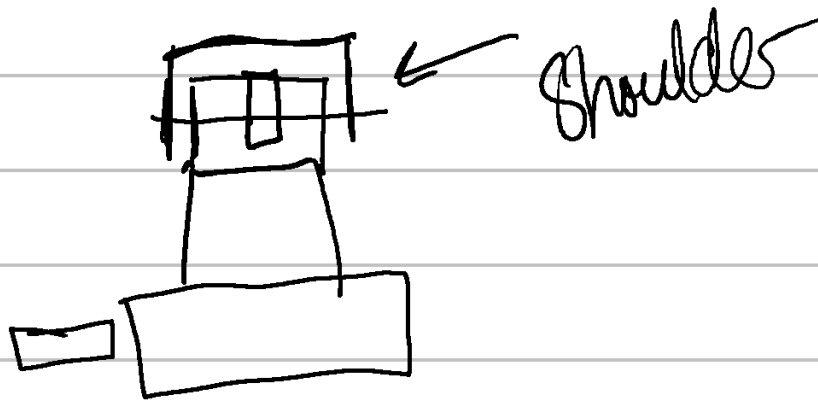
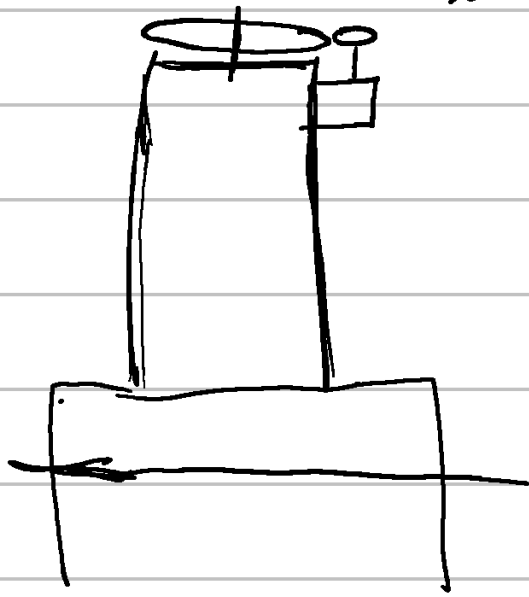


AS 5: Manipulation

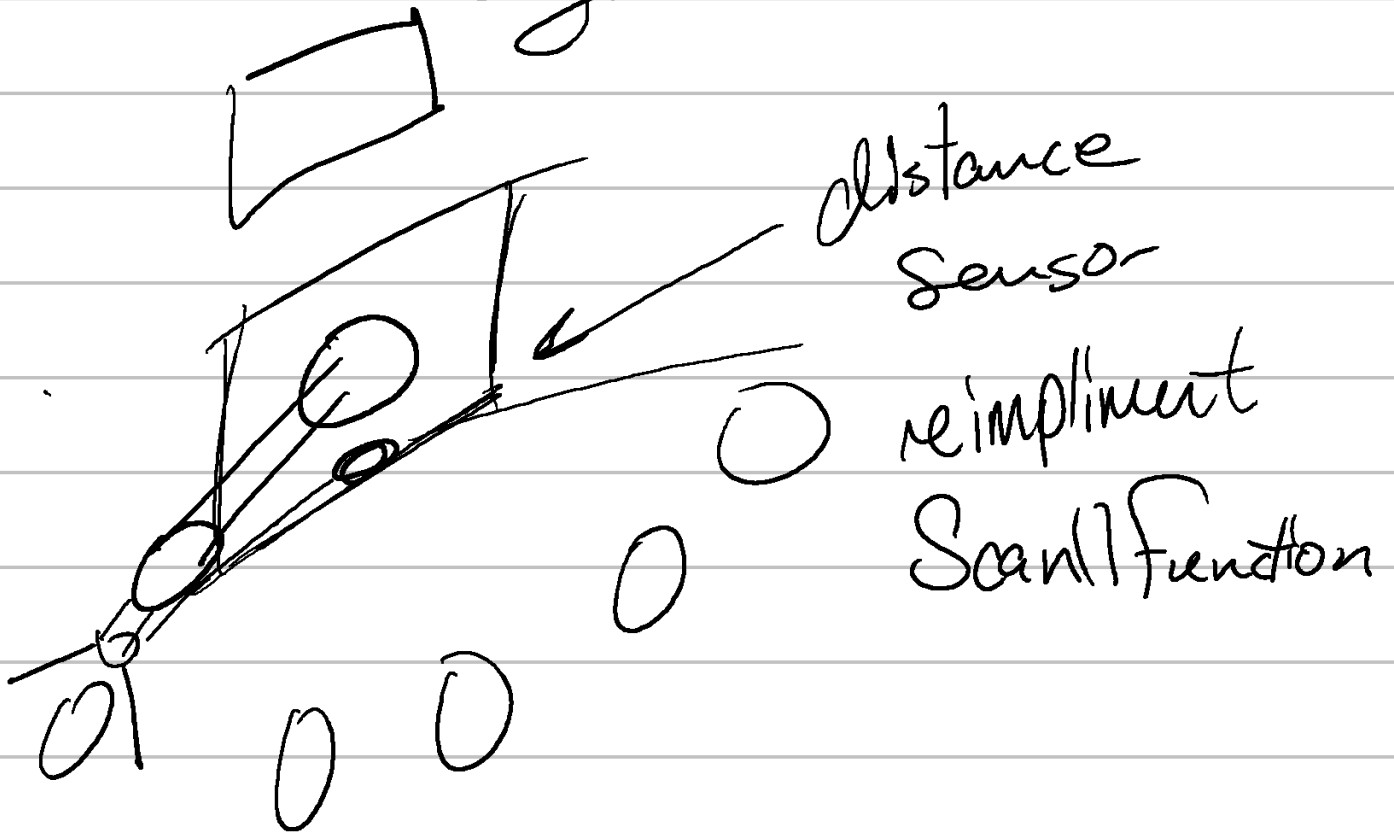






how do I
get this point
to not break

For the assignment

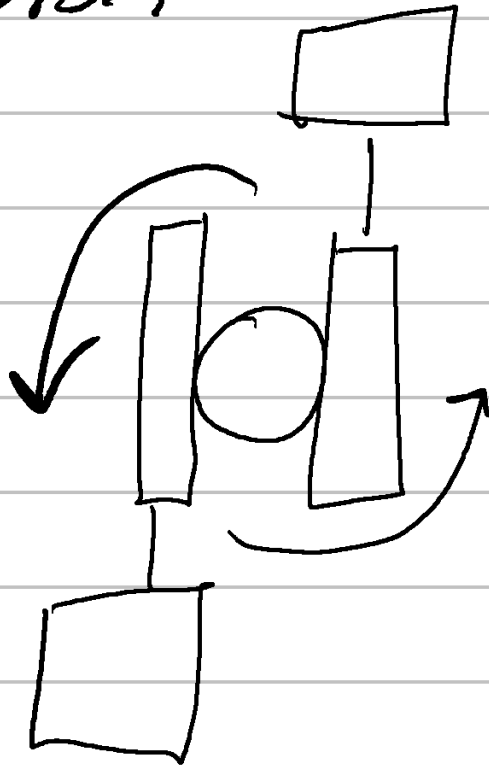




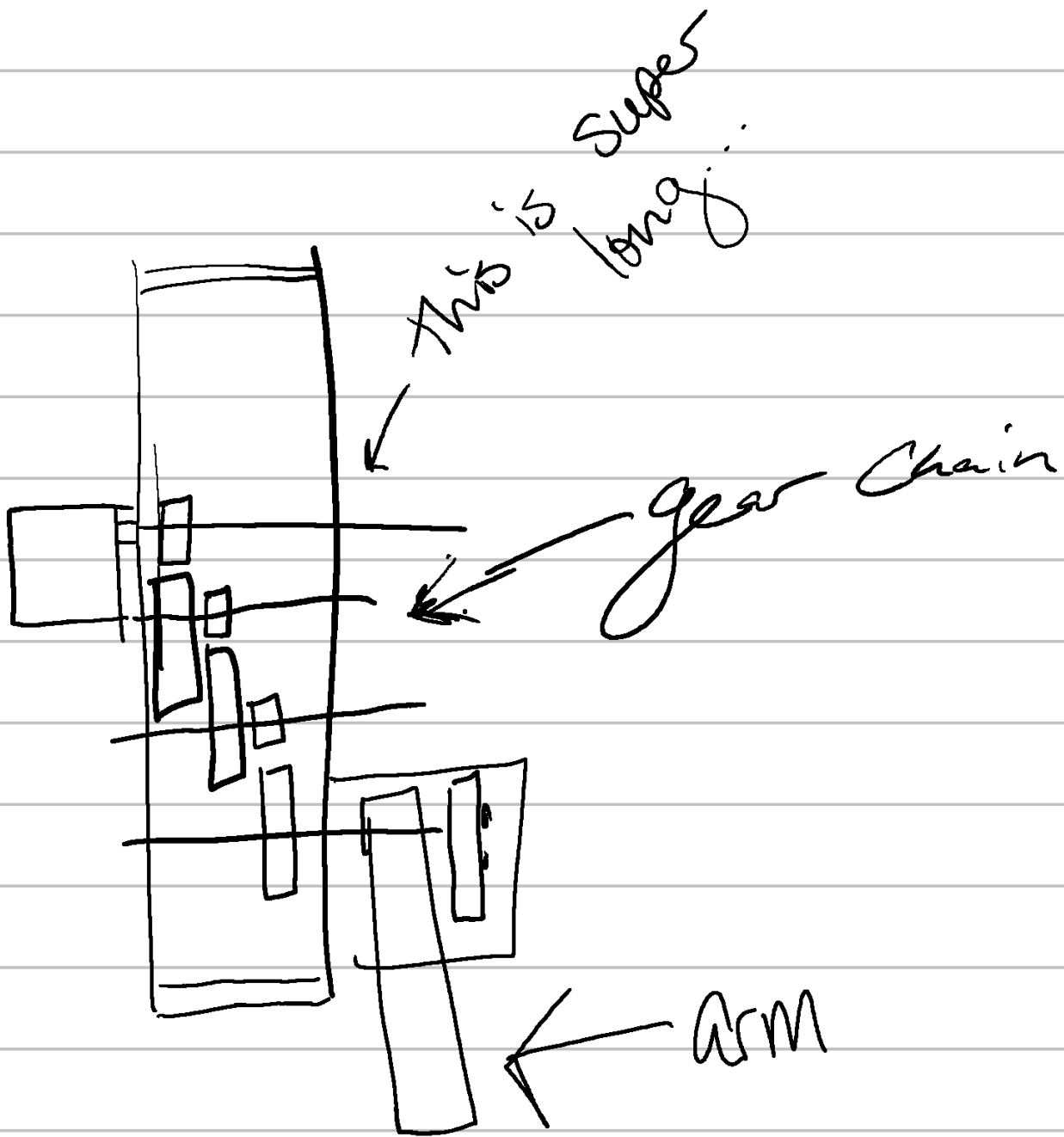
March 31, 2026

What needs done?:

We figured out how we
were gonna attach the
motors



April 3, 2026



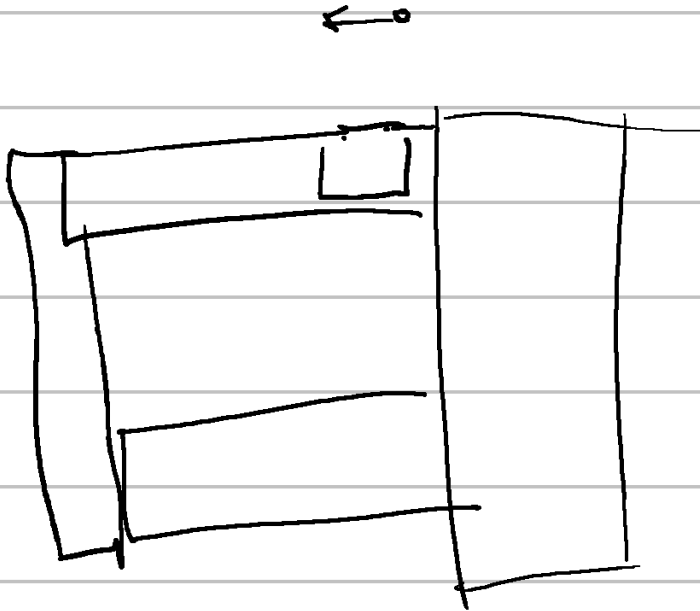
So this is a new design that was built w/o me here
So the problems of this new design are as follows:

- too much torque on the axle resulting in hardware failure
- Unstable design in terms of weight resulting in tip over.
- Screws exposed on bottom resulted in further instability



← a forementioned hardware failure

We are moving the center
of gravity



This way we are creating more
surface area for the weight
allowing it to remain stable.

Further issues and fixes

- I: Gear enclosure bowing
- F: Put additional supports on the bottom to pull the metal together
- I: Axel not inserting due to not being level.
- F: Added screws for variable level control as well as washers for arm to be level.
- I: Arm has too much play meaning it will jump a few in. when not told to. This will lead in the future to bad control.
- F: None leave and use code to account

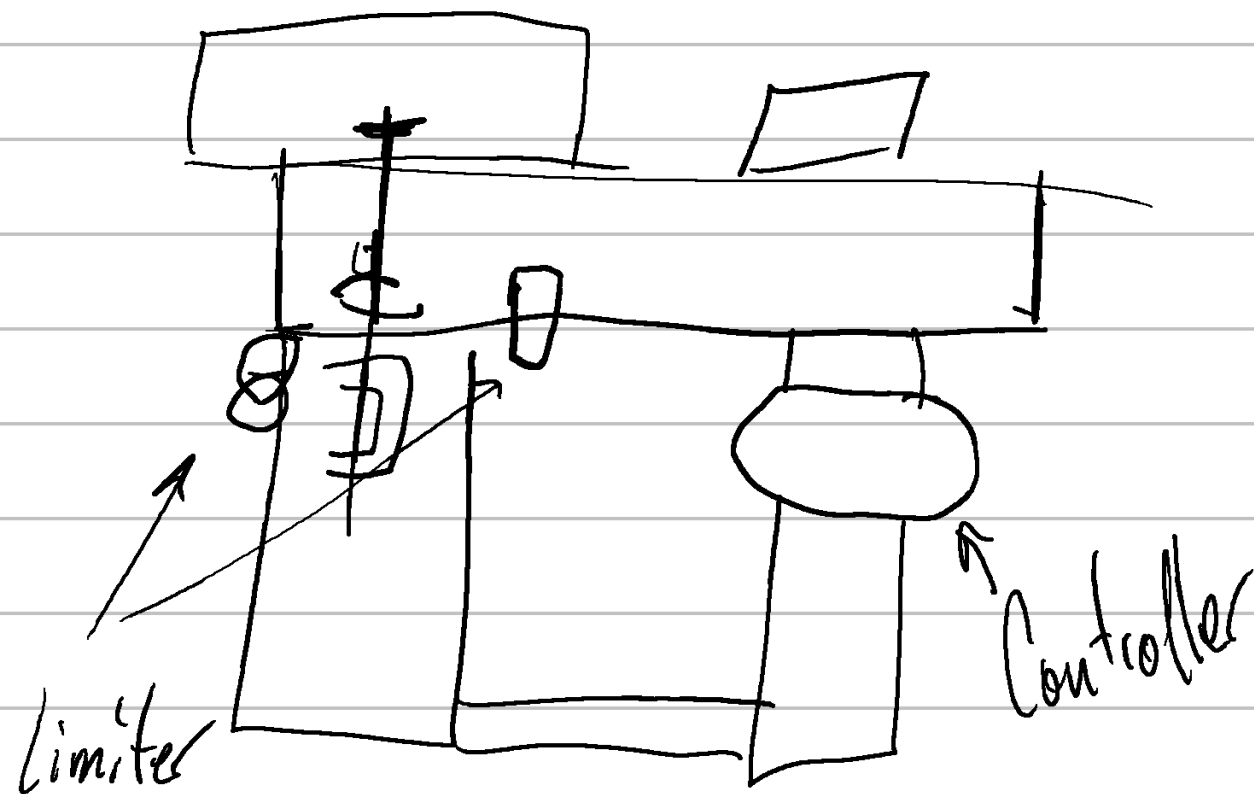
April 9th

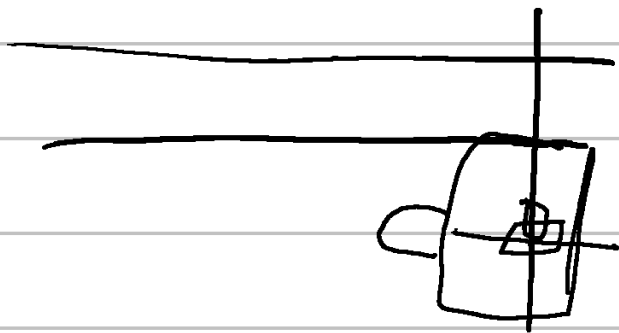
Since last time
Code has been done to
move the arm

has a ton of torque.
Can lift a lot.

Next step. End axis and more
code.

Base has 1 to 125 Ratio





We are attempting to get the design built

The main issue is stripped shaft collars

We are currently dealing
with a bobble on the
worm gears leading to
the gears disconnecting.

Apr. 14

We are starting coding
currently we are working
through the motors.

They appear to work well.

However due to gearing the
motors multiply error.

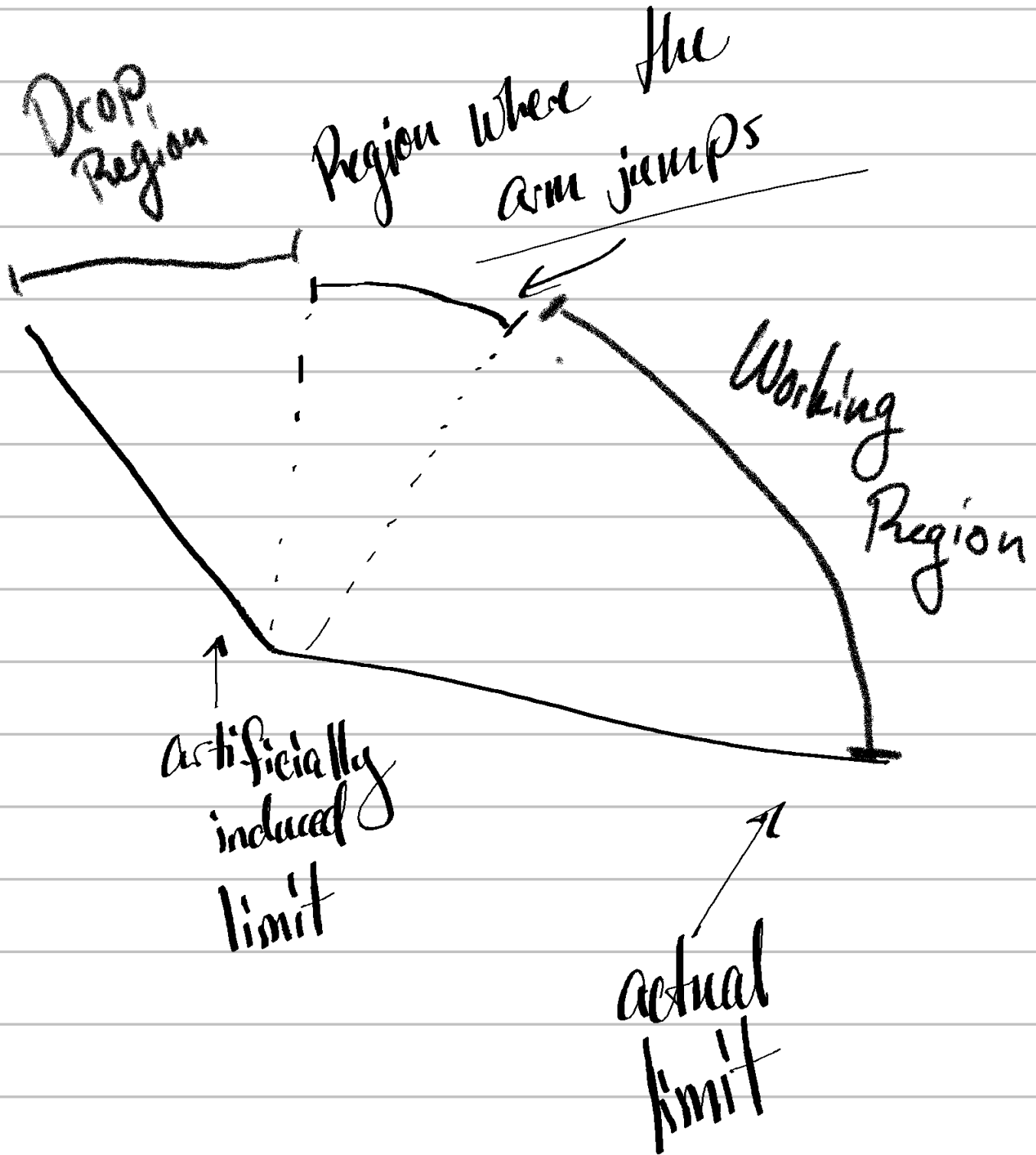
So we are going to see how
we can make this issue less
a problem.

We will do this by simply ignoring it... truly a life lesson is to be learned here.

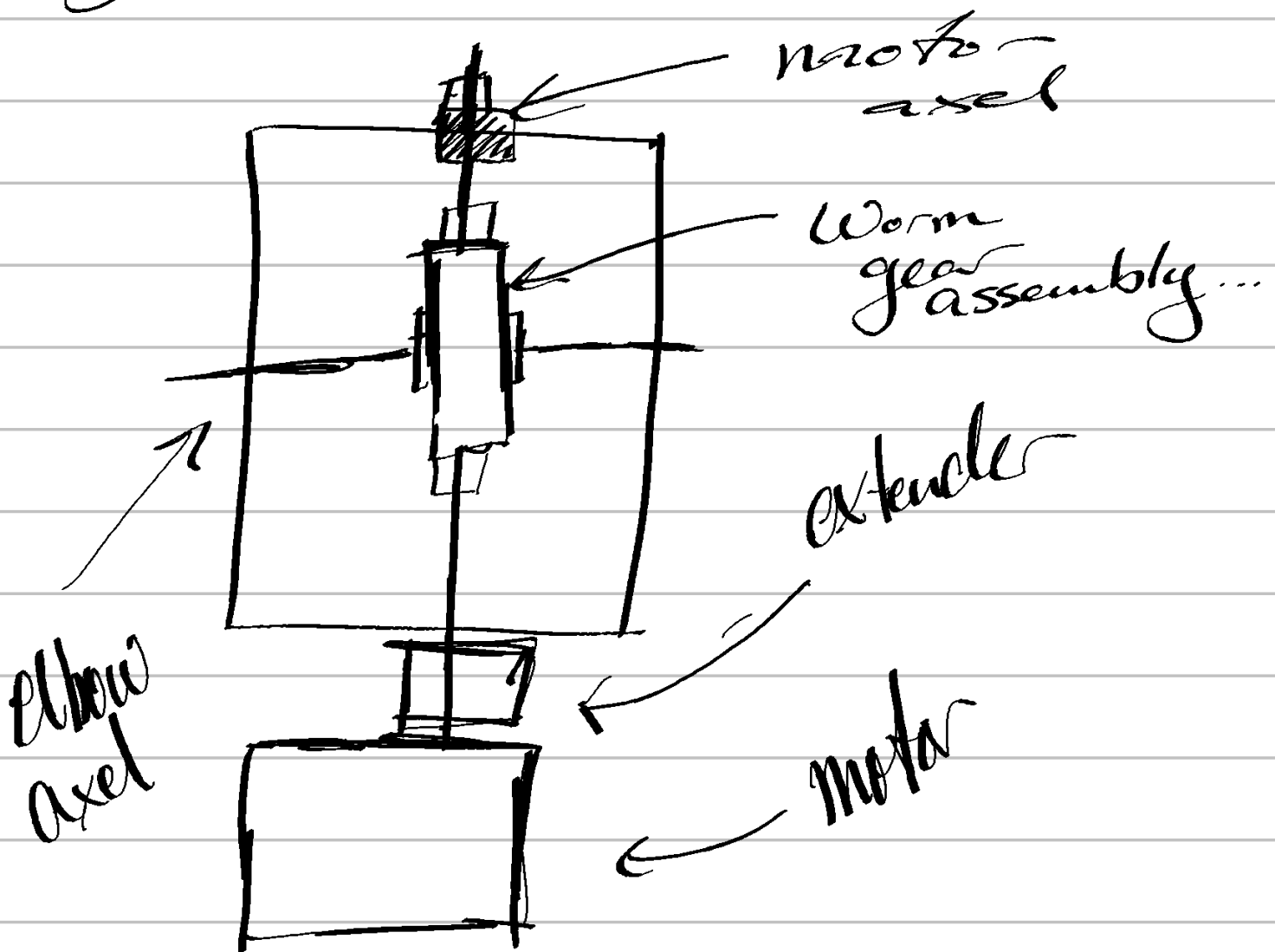
We will use the potentiometer to see where the arm is. This will cause us to be able to determine where the arm is.

Further, we will limit the arm to a set region to operate to allow this to be ignored

Jumping Issue



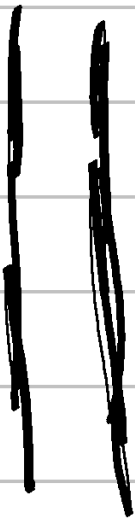
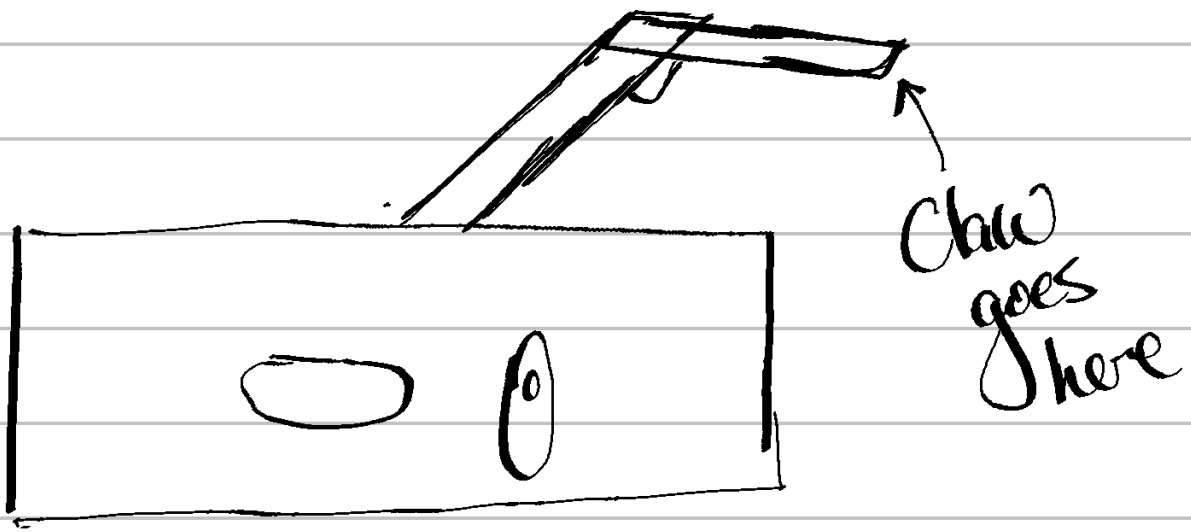
Another issue is that the axle which will force itself out of the motor due to the weight. This has caused issues where the axle will lift the worm gear out of place.



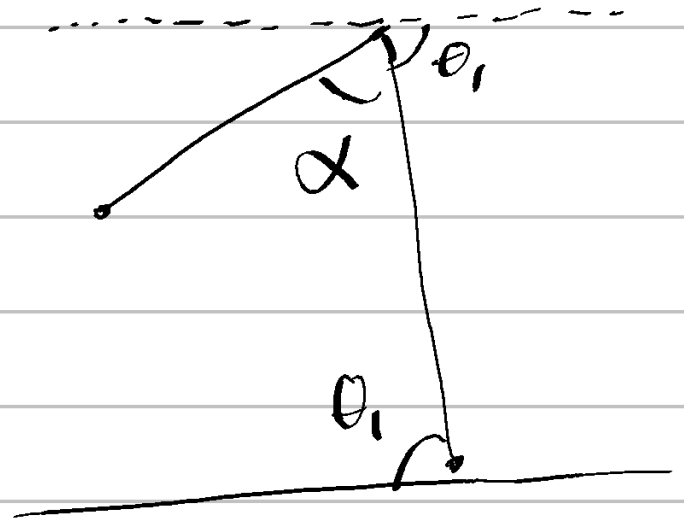
Apr. 16

We did some work out of class with the code in order to get started on the actual coding.

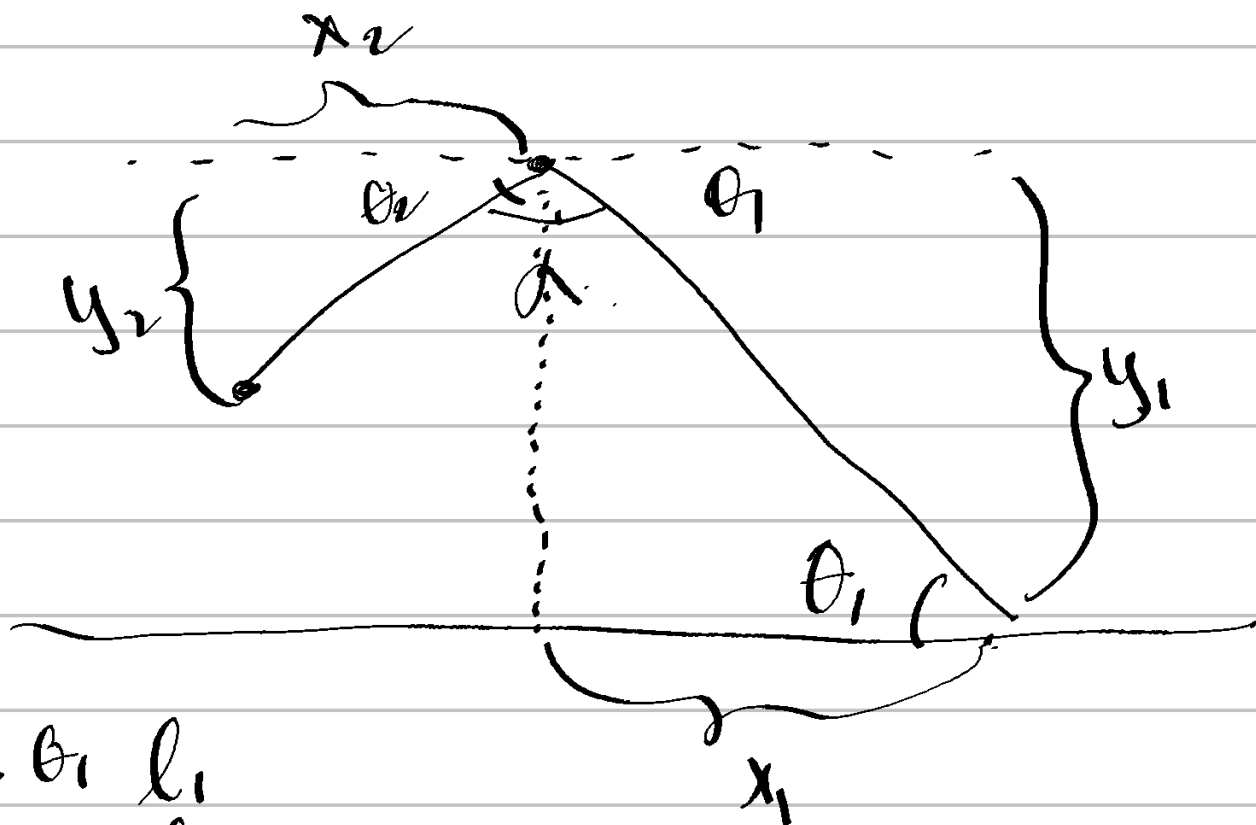
- Eric on his down time solved the equations out. We are working on implementation.



if distance $< l_1$
use l_2



$$180 - \theta_1 - \theta_2 = \alpha$$



$$x_1 = \cos \theta_1 l_1$$

$$y_1 = \sin \theta_1 l_1$$

if $\alpha > 180 - \theta_1$
then $y_1 + y_2$

if $\alpha > 180 - \theta_1$ else

$$\text{then } y = y_1 + y_2$$

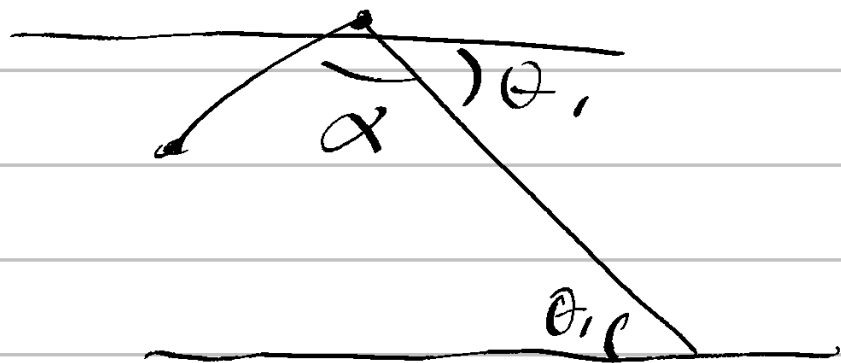
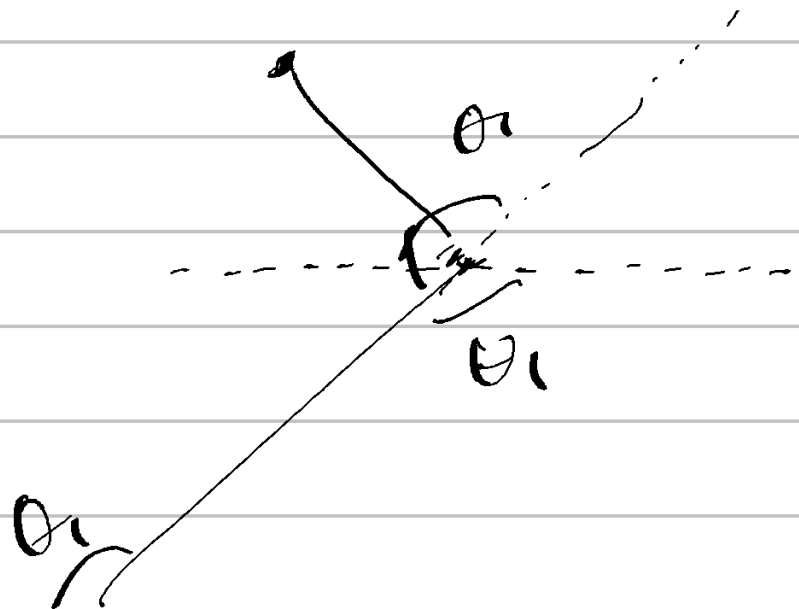
$$y = y_1 - y_2$$

if $\theta_1 < 90$

$$-(x_1 + x_2)$$

else

$$x_1 + x_2$$



$$x_1 = \cos \theta_1 l_1, y_1 = \sin \theta_1 l_1, x_2 = \cos \theta_2 l_2, y_2 = \sin \theta_2 l_2$$

Found an inf loop in
the code removed.
in the get-a-car

There is an issue where
our stop() keeps getting
overridden.

- So the issue is two threads
running at the same time.

The solution to this is to
have methods handle this
instead of trying to run two
threads at once.

Friday I plan on going through the
code and adding these methods
to handle logic.

- Get_into_tolerance ()

- Will find the tolerances
- Get robot into shape.

- We are going to add the limit switch safeties into our main loop as get_into_tolerance should normal use of switches.

- Finch_can ()

- This we will have the dist. sensor take a reading and pass it to the math.

= Retrieve_can ()

- This will take the output of the math and tell the arm to move.

The find and retrieve
are separate is for
safety in case one
method is broken we
can spot the error.

4/23/26

- Currently still locked on kinematics. Attempting to resolve this.
- This is going really slow due to the sheer amount of torque on the shoulder.
 - So the speed is currently being worked on.
 - Need to get the C parallel to the table
 - having key components put into methods for debugging.

- We have the arm working.
now we are adding weight to get approved.
- Had to lube the gears so it would run more smoothly
- The angles should pre-check if the point will hit.
- We do that with the Hypotenuse